

Programming Fundamentals Lab

Final Project

Connect 4 Game

Code:

```
#include <iostream>
#include <string>
#include <ctime>
#include <fstream>
std::string player1, player2;
int turns = 0;
char ch;
int r = 0, b = 0;
char play_opt();
char option[50];
const int ROWS = 6;
const int COLUMNS = 7;
char board[ROWS][COLUMNS];
void file();
std::string fill;
// to initialize board
void initializeBoard()
{
    for (int i = 0; i < ROWS; i++)
    {
        for (int j = 0; j < COLUMNS; j++)
        {
            board[i][j] = '.';
        }
    }
}
// to print board
```

```

void printBoard() {
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLUMNS; j++) {
            std::cout << board[i][j] << " ";
        }
        std::cout << std::endl;
    }
}

//storing moves
bool placeDisc(int column, char disc) {
    if (column < 1 || column > COLUMNS) {
        return false;
    }
    //storing moves from the end
    for (int i = ROWS - 1; i >= 0; i--)
    {
        if (board[i][column - 1] == '.')
        {
            board[i][column - 1] = disc;
            return true;
        }
    }
    return false;
}

//horizontal
bool checkWin(char disc)
{
    for (int i = 0; i < ROWS; i++)
    {
        for (int j = 0; j < COLUMNS - 3; j++)
        {
            if (board[i][j] == disc && board[i][j + 1] == disc
&&
                board[i][j + 2] == disc && board[i][j + 3] ==
disc)
            {
                return true;
            }
        }
    }
    //vertical checking
    for (int i = 0; i < ROWS - 3; i++)
    {
        for (int j = 0; j < COLUMNS; j++)
        {

```

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        if (board[i][j] == disc && board[i + 1][j] == disc
&&
        board[i + 2][j] == disc && board[i + 3][j] ==
disc)
        {
            return true;
        }
    }
}
//right to left diagonal
for (int i = 3; i < ROWS; i++) {
    for (int j = 0; j < COLUMNS - 3; j++) {
        if (board[i][j] == disc && board[i - 1][j + 1] ==
disc &&
        board[i - 2][j + 2] == disc && board[i - 3][j
+ 3] == disc) {
            return true;
        }
    }
}
//lef to right diagonal
for (int i = 3; i < ROWS; i++) {
    for (int j = 3; j < COLUMNS; j++) {
        if (board[i][j] == disc && board[i - 1][j - 1] ==
disc &&
        board[i - 2][j - 2] == disc && board[i - 3][j
- 3] == disc) {
            return true;
        }
    }
}
return false;
}
//draw
bool checkDraw() {
    for (int i = 0; i < COLUMNS; i++) {
        if (board[0][i] == '.') {
            return false;
        }
    }
    return true;
}
void all()
{
    //instruction option
    std::string ins = "instructions";
    std::string inst = "Instructions";

```

```

    if (option == ins || option == inst)
    {
        std::cout << "\t\t\tINSTRUCTIONS:";
        std::cout << "\n\nThe Connect 4 game is a classic strategy
game in which 2 players go head-to-head in a battle to own the
grid!\n";
        std::cout << "Players choose blue or red discs. \nThey
drop the discs into the grid, starting in the middle or at the
edge to stack their colored discs\nupwards, horizontally, or
diagonally.\nUse strategy to block opponents while aiming to be
the first player to get 4 in a row to win. ";
    }
    //credits option
    std::string cr = "credits";
    std::string cred = "Credits";
    if (option == cr || option == cred)
    {
        std::cout << " \t\t\t CREDITS:\nProgrammed by: \nAyesha
Noor\nHashim Shaid\nAhmad Abdullah ";
    }

    //play option
    std::string pl = "play";
    std::string play = "Play";
    if (option == pl || option == play)
    {
        turns = 0;
        ch = play_opt();
    }
    //records option
    std::string re = "records";
    std::string rec = "Records";
    if (option == re || option == rec)
    {
        file();
    }
    //exit game
    std::string ex = "exit";
    std::string exit = "Exit";
    if (option == ex || option == exit)
    {
        std::exit(0);
    }
}
//all play criteria and code
char play_opt() {

```

```

std::cout << "Enter name for Player 1 (Red): ";
std::cin >> player1;
std::cout << "Enter name for Player 2 (Blue): ";
std::cin >> player2;
std::cout << "\n\t For Toss select head or tail\n " <<
player1 << " enter choice:";
std::string currentPlayer;
std::string choice;
std::cin >> choice;
srand((int)time(0));
// Toss by random numbers
int time = (rand() % 2) + 1;
if (time == 1)
{
    if (choice == "head" || choice == "Head")
    {
        std::cout << " head!  " << player1 << " wins the
race";
        std::cout << " wins the race.";
    }
    else
    {
        std::cout << " tail!  " << player1 << " wins the
race";
        std::cout << " wins the race.";
    }
    currentPlayer = player1;
}
if (time == 2)
{
    if (choice == "head" || choice == "Head")
    {
        std::cout << " tail!  " << player2 << " wins the
race";
        std::cout << " wins the race.";
    }
    else
    {
        std::cout << " head!  " << player2 << " wins the
race";
        std::cout << " wins the race.";
    }
    currentPlayer = player2;
}
std::cout << std::endl;

```

```

initializeBoard();//calling function of initialize board
printBoard();// to print board
char currentDisc = 'R';
int column;
//storing moves and displaing turns
while (true) {
    std::cout << currentPlayer << "'s turn. Please enter
your choice (column 1-7): ";
    std::cin >> column;
    if (!placeDisc(column, currentDisc)) {
        std::cout << "Invalid move! Try again." <<
std::endl;
        continue;
    }
    turns++;
    printBoard();
    //checking winning conditions
    if (checkWin(currentDisc))
    {
        std::cout << currentPlayer << " wins in " << turns
<< " turns!" << std::endl;
        break;
    }
    // draw case
    if (checkDraw())
    {
        std::cout << "Game is a draw!" << std::endl;
        break;
    }

    currentPlayer = (currentPlayer == player1) ? player2 :
player1;
    currentDisc = (currentDisc == 'R') ? 'B' : 'R';
}
fill = currentPlayer;
return currentDisc;
}
int main()
{
    for (int i = 0; i >= 0; i++)
    {
        //menu
        std::cout << "Menu:\n\t\t      Play\n\tInstructions\t\t
Credits\n\tRecords\t\t\t\t Exit\n";
        //getting option from menu from user

```

```

        std::cout << "\nEnter option: ";
        std::cin.getline(option, 50);
        all();
    }
    system("pause");
}
//file handling
void file() {
    std::ofstream out;
    std::ifstream in;
    out.open("file.txt");
    if (!out.is_open())
    {
        std::cout << "error in opening file";
    }
    else
    {
        out << fill << "\t\t" << " wins in " << turns << "
turns!" << std::endl;
    }
    out.close();
    in.open("file.txt");
    if (!in.is_open())
    {
        std::cout << "error in opening file";
    }
    else
    {
        std::string line;
        char ch;
        while (!in.eof())
        {
            in.get(ch);
            std::cout << ch;
        }
    }
}
}

```

Output:

```
Menu:
    Play
Instructions   Credits
Records       Exit

Enter option: Instructions
INSTRUCTIONS:
The Connect 4 game is a classic strategy game in which 2 players go head-to-head in a battle to own the grid!
Players choose blue or red discs.
They drop the discs into the grid, starting in the middle or at the edge to stack their colored discs
upwards, horizontally, or diagonally.
Use strategy to block opponents while aiming to be the first player to get 4 in a row to win. Menu:
    Play
Instructions   Credits
Records       Exit

Enter option: credits
CREDITS:
Programmed by:
Ayesha Noor
Hashim Shaïd
Ahmad Abdullah Menu:
    Play
Instructions   Credits
Records       Exit

Enter option: play
Enter name for Player 1 (Red): Ahmad
Enter name for Player 2 (Blue): Ali

For Toss select head or tail
Ahmad enter choice:1
head! Ali wins the race wins the race.
. . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . . .
Ali's turn. Please enter your choice (column 1-7): 4
. . . . .
. . . . .
. . . . .
. . . . .
. . R . . .
Ahmad's turn. Please enter your choice (column 1-7): 5
. . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . R B . .
Ali's turn. Please enter your choice (column 1-7): 4
. . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . R B . .
Ahmad's turn. Please enter your choice (column 1-7): 5
. . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . R B . .
. . . R B . .
Ali's turn. Please enter your choice (column 1-7): 4
. . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . R B . .
. . . R B . .
Ahmad's turn. Please enter your choice (column 1-7): 5
. . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . R B . .
. . . R B . .
Ali's turn. Please enter your choice (column 1-7): 4
```



```
. . . . .
. . . R . . .
. . . R B . .
. . . R B . .
. . . R B . .
Ali wins in 7 turns!
Menu:
      Instructions      Play      Credits
      Records          Exit
Enter option: Menu:
      Instructions      Play      Credits
      Records          Exit
Enter option: Records
Ali wins in 7 turns!
Menu:
      Instructions      Play      Credits
      Records          Exit
Enter option: Exit

C:\Users>window 10\Downloads\FinalProject\x64\Debug\FinalProject.exe (process 17376) exited with code 0.
Press any key to close this window . . .
```

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